

**A Unified Machine Learning Framework for Customer Segmentation and Product
Preference Prediction in Retail Analytics**

Priyal Rawat

DeVos Graduate School, Northwood University

MGT 665: Solving Business Problems W/ Machine Learning

Dr. Itauma Itauma

07/11/2025

Abstract

This study offers a comprehensive machine learning method for customer segmentation and product purchase prediction, utilizing a real-world retail marketing dataset from Kaggle combining customer demographics, purchasing behavior, and marketing campaign responses. Various unsupervised learning models such as K-Means, Agglomerative Clustering, and DBSCAN were performed before and after various forms of dimensionality reduction using Principal Component Analysis (PCA). Following PCA K-Means performed better with K-Means achieving a silhouette score of 0.38, compared to 0.21 before PCA, and showed improved clustering separability relating to customer groups with ten targets identified that could be interpreted easily. DBSCAN didn't work out after PCA. In the supervised modelling phase, five classification models were built to predict what the next product a customer would purchase. Of the models, XGBoost had the highest performance with an accuracy of 0.77 and F1 scores of 0.75 (after SMOTE). Hyperparameter tuning and SMOTE was applied but there was no significant improvement in the model performance. The performance of the models was not at par and relevant for real world settings. Cluster labels and product prediction outputs were presented at the same time, giving customers insights into what products could be considered next, thus better personalizing their profile. The predictive improvements detailed in this study were limited, however, this study has demonstrated that retail businesses can substantially benefit by integrating unsupervised and supervised modelling outputs to influence, better inform, customer data-driven retail strategies. The performance of the models was not at par and relevant for real world settings. Future work includes integrating deep learning models.

Introduction

In today's data-driven retail environment, understanding customers' behavior is one of the most important and strategic requirements for businesses. Customer market has become more complex than ever before and personalized customer experience is now treated as a standard services, retailers have to go further and beyond to retain customers and provide best customer experience and for that understanding consumer purchasing behaviors is extremely important and plays a major role. This is the most promising area for growth is the ability to understand customer behavior predicting what a customer is most likely to buy. Some retailers and brands have traditionally relied on gut instinct or manual observations. However, machine learning models can analyze the large amount of customer data, their behavioral, demographic data produced since the first interaction and provide meaningful insights for the businesses.

Traditional segmentation models have guided marketing efforts for the longest period of time, but they usually rely on static attributes or rely on outdated assumptions. This field has made considerable advances with the integration of behavioral segmentation (Ramkumar et al., 2025; Suh, 2025) with frameworks for predictive modeling of loyalty, churn, or customer value (Kabir et al., 2025; Bhatia & Gupta, 2025). Still, there are very few studies where the focus on unsupervised customer segmentation is linked to predicting customer product preferences that too from a structured dataset which has multiple product categories and marketing touch points that tracks spending behavior of a customer.

To bridge this gap, this study focuses on a unified machine learning approach that integrates unsupervised customer segmentation with supervised product preference prediction, based on a real-world retail dataset. The study starts by grouping customers together using clustering techniques such as K-Means, Hierarchical clustering and DBSCAN and these models

have been evaluated using metrics such as Silhouette Score, WCSS, Davies-Bouldin Index and Calinski-Harabasz (Deldadehasl et al, 2025; Lathifah & Azzahra, 2025). Principal component analysis (PCA) has been applied to further improve the segmentation. The clusters formed from an unsupervised learning are then used as extra inputs for classification algorithms such as logistic regression, Knn, decision tree, random forest, XGboost to predict what customers is going to buy in their next purchase (Kabir et al, 2025; Patel & Mehta, 2024). Through this approach, model will be able to predict customer's product preference with high accuracy but also segment customer into different clusters for better service and effective marketing strategy.

Literature Review

Many studies have used machine learning techniques for customer segmentation, prediction and classification tasks which shows progress in both methods and practical applications.

Recent studies shows the trend of using machine learning methods to predict customer segmentation and behavior. Suh's (2025) research was an analysis of behavioral data for 40,000 plus users for the home appliance industry and used clustering techniques to identify unique customer types of "information seekers" and "experience-oriented" buyers. The study physically showed how unique customer types develop at different customer journey stages and proposes there are future implications through personalized marketing directly but also acts as utilization of proactive customer relationship management strategies.

Bhatia and Gupta (2025) examined physical retail, and they utilized predictive analytics, supervised models like linear regression, and Random forest, to forecast monthly sales and purchasing behavior. The findings of the data spell out that transactional and demographic data was useful to forecast, and in some respects, considering a more fine-grained approach, such as

looking at streaming data would provide the researchers with the best data to understand what is happening with customer dynamics over time.

There are researchers who have explored more complex clustering models. Nasirzonouzi (2025) used network science to graph the connections between customer and product to create greater possibilities for segmentation. This approach outperformed static clustering, and he plans to do further work to evaluate some of the more recent centrality measures and implement the approach in other industries. Likewise, Ramkumar et al. (2025) adjusted the traditional RFM model by incorporating K-Means. The research showed customer segments like "frequent small spenders" and "infrequent high-value buyers." As a result, retailers improved their campaign performance by 15%.

Similarly, other researchers employed clustering using analytical techniques. Deldadehasl et al. (2025) used clustering with a modified RMD model to cluster hotel customers and used TOPSIS and Customer Lifetime Value to rank these clusters based on customer needs and concerns. In their analysis, they employed association rule mining to identify clusters of customers which had similar base behavior profiles and extracted behavioral patterns. Lathifah and Azzahra (2025) also used K-Means and PCA to cluster retail customers on demographic and behavioral data, which showed a relatively high Davies-Bouldin score at ~ 1.27 that assisted in targeting loyalty and promotion campaigns accurately. Awaliyah et al. (2024) applied these techniques in the online retail context, and they also concluded that better targeting these segments leads to a 12% increase in overall marketing results.

Several papers have made a move towards beyond identification and segmentation. Dhanushkodi et al. (2024) created a full data mining model of supermarkets which included RFM, SVM, and Neural Networks achieved an F1-score of 0.97, they also used association rules

to pull say recommendations for helpful insights. Finally, Kabir et al. (2025) implemented a combination of clustering models and classification models (SVM, Random Forest and Gradient Boosting) on e-commerce data and achieved high precision-recall rates (66-75%) with the aforementioned models, they even stated that their deep learning models (ANN and CNN), fared even better with 90%+ accuracy. They advocate for future research to consider time-based models and using explainable AI.

Most of the studies focus on traditional machine learning and deep learning for segmentation and targeting have focused on general classification tasks, like churn, loyalty tier, or general expressions of interest in a product, and fail to utilize the entirety of the behavioral richness of transaction patterns in multiple product categories.

Majority of the studies focused on using machine learning and deep learning for segmentation, customer churn, product recommendation, but very few study focused on rich customer behavior along with transaction patterns in multiple product categories. Moreover, many approaches rely on a limited feature set or using behavioral summaries like RFM and do not generally focus on predicting first purchase behavior. This study focuses on bridging the gap by using a complete set of retail data with real customer spending, various marketing channel usage and campaign response features for identification, segmentation and multi-class classification of predicting product preferences based only on initial customer transaction characteristics. By applying segmentation clustering and multi-class classification methods, this research will improve the potential for better segmentation of customers and targeted marketing efforts for more personalized product offerings and stronger ROI through improved marketing spend.

Methodology

Research Design

This research utilizes a structured machine learning process to explore customer behavior and predict product preference in retail. This process begins using unsupervised learning techniques such as K-Means, Hierarchical Clustering, and DBSCAN for customer segmentation. Principal Component Analysis (PCA) has been applied and evaluated using methods like Silhouette Score, WCSS, Davies-Bouldin Index, and Calinski-Harabasz for better and effective segmentation. After creating cluster labels for customers, supervised learning is used to predict which product each customer will order. Five machine learning models were built which are Logistic Regression, k-NN, Decision Tree, Random Forest, and XGBoost, and hyperparameter tuning and SMOTE was done and evaluated using Accuracy, Precision, Recall, F1 score. This study focuses on predicting product customers along with customer segmentation which can provide meaningful actionable insight for the retail businesses.

Dataset description

The dataset used in this study is downloaded from Kaggle which has 29 variables of a 2240 customers from a retail company representing demographic information such as birth year, level of education, marital status, and income, as well as household composition like numbers of kids and number of teenagers, as well as customer tenure which is represented by Dt_Customer. It also has behavioral information like Recency meaning days since last purchase. Responses to different marketing campaigns like AcceptedCmp1 to AcceptedCmp5, and with the last field being Response.

The dataset also has demographic information along with rich insights on how much customers spent over six product categories which are wines, fruits, meats, fish, sweets, and gold. It even records how many purchases were made through deals, online, catalog, and in-store purchases and how often the shopper visited the website. The variety of characteristics and mix of demographics and behavioral variables provides detailed insights into the lifestyle, preferences, and responsiveness of each customer which could be usefully analyzed or segmented and could also be useful for predictive modelling.

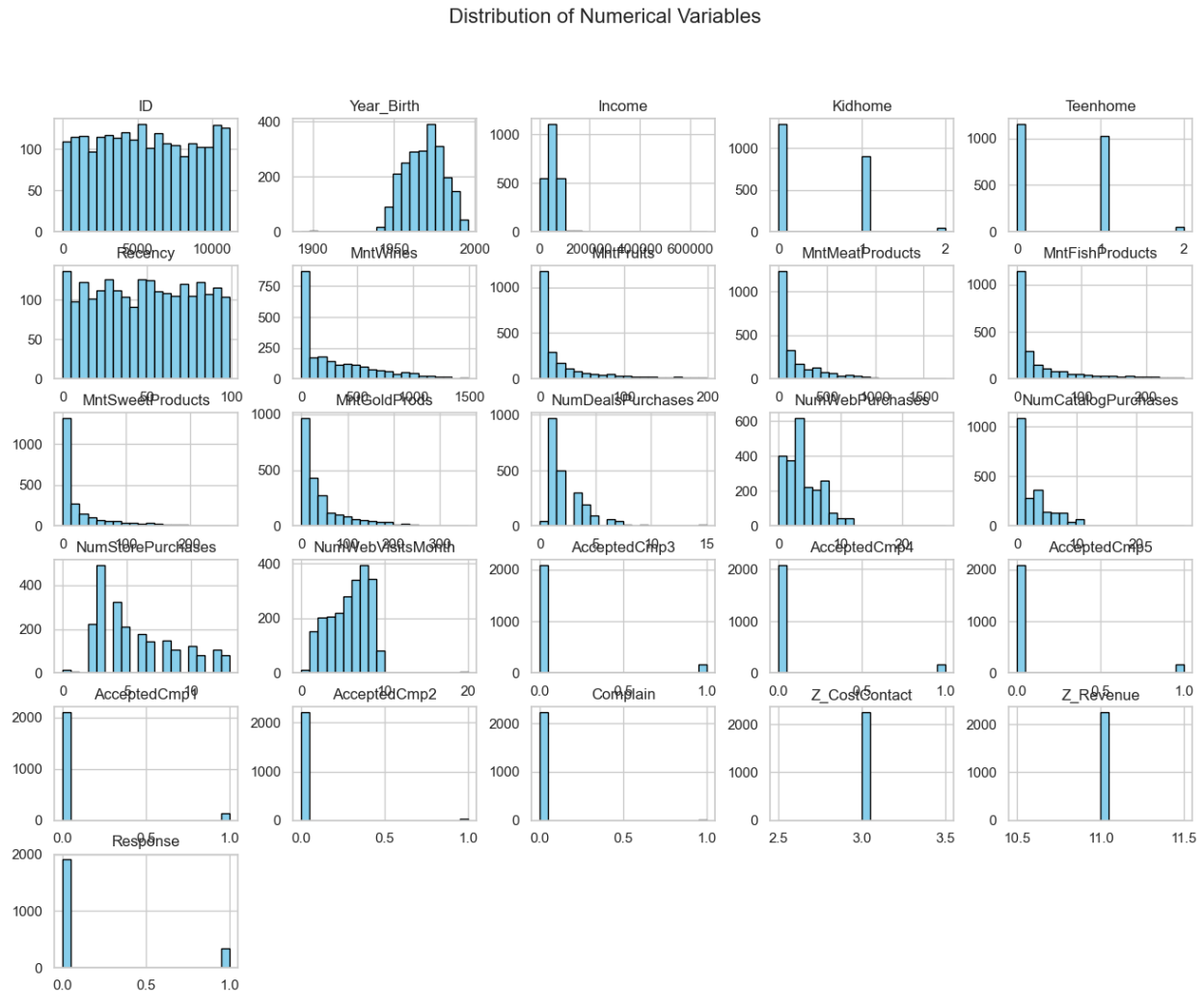
Details of Features from Kaggle are as below:

Year_Birth: The birth year of the individual. Education: The highest level of education attained by the individual. Marital_Status: The marital status of the individual. Income: The annual income of the individual. Kidhome: The number of young children in the household. Teenhome: The number of teenagers in the household. Dt_Customer: The date when the customer was first enrolled or became a part of the company's database. Recency: The number of days since the last purchase or interaction. MntWines: The amount spent on wines. MntFruits: The amount spent on fruits. MntMeatProducts: The amount spent on meat products. MntFishProducts: The amount spent on fish products. MntSweetProducts: The amount spent on sweet products. MntGoldProds: The amount spent on gold products. NumDealsPurchases: The number of purchases made with a discount or as part of a deal. NumWebPurchases: The number of purchases made through the company's website. NumCatalogPurchases: The number of purchases made through catalogs. NumStorePurchases: The number of purchases made in physical stores. NumWebVisitsMonth: The number of visits to the company's website in a month. AcceptedCmp3: Binary indicator (1 or 0) whether the individual accepted the third marketing campaign. AcceptedCmp4: Binary indicator (1 or 0) whether the individual accepted

the fourth marketing campaign. AcceptedCmp5: Binary indicator (1 or 0) whether the individual accepted the fifth marketing campaign. AcceptedCmp1: Binary indicator (1 or 0) whether the individual accepted the first marketing campaign. AcceptedCmp2: Binary indicator (1 or 0) whether the individual accepted the second marketing campaign. Complain: Binary indicator (1 or 0) whether the individual has made a complaint. Z_CostContact: A constant cost associated with contacting a customer. Z_Revenue: A constant revenue associated with a successful campaign response. Response: Binary indicator (1 or 0) whether the individual responded to the marketing campaign.

Exploratory Data Analysis & Data Preprocessing

In this part of the study, detailed examination was done of customer attributes including demographics, behaviors, and purchasing activities to understand and find patterns within the data. Distribution of important numerical variables, which included income, spending amounts in product categories, and recency of purchases, by using histograms and boxplots. Correlation heatmap, pair plot was used to understand the relationships of features and to explore multicollinearity, in which two or more independent variables are correlated as well as feature importance. Outliers were examined in the income variables, to understand for the model development. Further, categorical variables such as education level and marital status were analyzed using count plots. This EDA process allowed us to not only develop a good understanding of customer profiles but also guide useful feature selection and preprocessing steps with dimensionality reduction for use in the exploration for clustering and predictive modeling efforts.

Figure 1*Distribution of numerical variables*

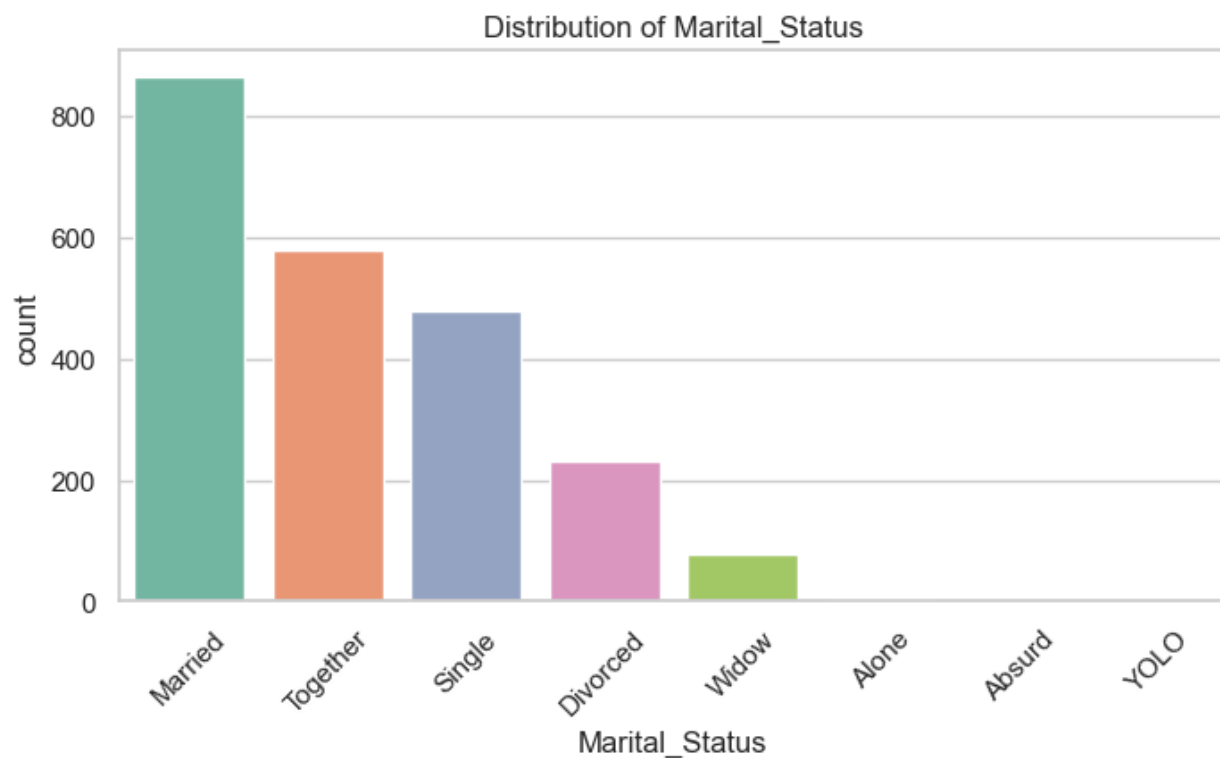
Note: Figure 1 has distribution of all numerical variables to identify skewness, outliers, and the likelihood each would need to be transformed. Most of the product spending and campaign acceptance variables were right-skewed. Various features had some features, such as $Z_CostContact$ and $Z_Revenue$, with no variance, indicating low predictive value.

Distribution of all of the numerical variables was done using histograms. The variables related to product spending such as $MntWines$, $MntMeatProducts$, $MntGoldProds$, etc. are

strongly right-skewed which suggests that most customers tend to spend less in products the companies sells, while only a few customers are responsible for a high volume of sales. This same skewed behavior also was noted in the variables related to purchases such as NumWebPurchases, NumCatalogPurchases, etc. and marketing campaign responses which are AcceptedCmp1 through AcceptedCmp5. As binary values like Response and Complain there was an imbalance with all of the values being concentrated around 0. There were two variables Z_CostContact and Z_Revenue that had values that were constant across all records, it seemed these variables do not have informational value so they were excluded in the preprocessing steps. These graphs helped in deciding on variable transformation, variable selection, and maximizing input for modeling.

Figure 2

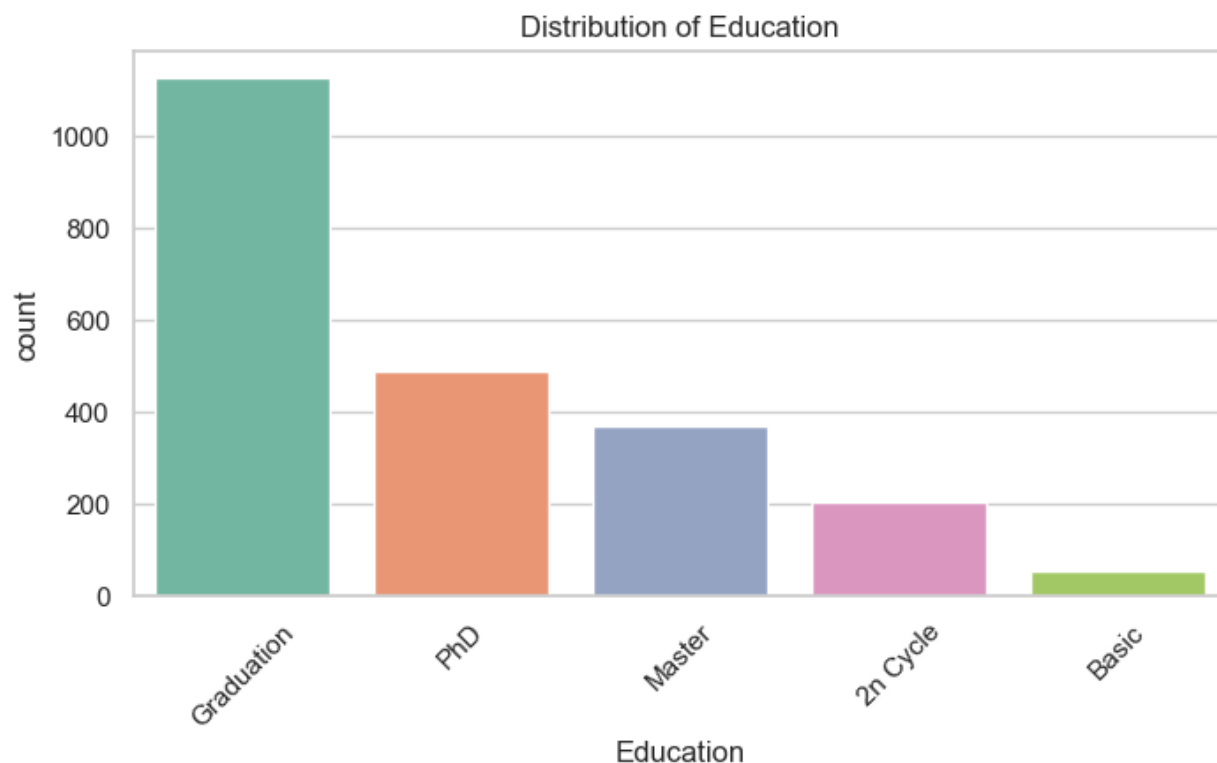
Distribution of Marital Status



Note: The overall depiction of the Marital_Status variable shows a majority of customers as Married, Together, or Single. Also, there are categories such as YOLO, Absurd, and Alone contain few customers worth noting to a total of 10 or less.

Figure 3

Distribution of Education



Note: The Education variable shows that the majority of customers have a Graduation degree, followed by PhD or Master's degree. The least number of customers are in the 2n Cycle and Basic categories. This disproportion supports a fairly educated customer base, but also can help inform marketing tactics based on their academic education as well.

Handling Missing Values

Table 1*Missing Values in the dataset*

Variable	Missing Values	Percentage (%)
Income	24	1.07

Note: Table 1 represents missing value in the dataset. It was identified that there was small number of missing value in the variable income. About 24 records were missing which accounts to about 1.07% of the dataset.

The proportion of missing values was minimal and since income is a continuous variable, missing values were handled using median imputation to manage overall distribution and bias from extreme values.

Variable Summary, Feature Engineering and Transformation

Table 2*Summary of Variable Type*

Type	Count	Variable Name
Numerical	26	ID, Year_Birth, Income, Kidhome, Teenhome, Recency, MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds, NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5, AcceptedCmp1, AcceptedCmp2, Complain, Z_CostContact, Z_Revenue, Response
Categorical	2	Education, Marital_Status
Date	1	Dt_Customer

Note: The dataset has 26 numerical variables, 3 categorical variables and 1 date variable.

Numerical features included Income, Recency, and spending on different products, and the

categorical variables included Education, Marital_Status, and Dt_Customer. The Dt_Customer variable was a date-type column and included the enrollment date of customers.

To enhance dataset value and to improve model performance, several feature engineering steps were implemented. First an Age variable was derived from the Year_Birth variable by subtracting Year_Birth from 2025 where 2025 is the age of the customer today. Then, children was made from summing up both Kidhome and Teenhome which gives us a total for children in a household. Total_Spent was created by summing six previously created spending variables which are MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, and MntGoldProds, it is to quantify each customers overall purchasing behavior. Yellow, Absurd and Alone values in the Marital_Status variable were put into a single category called Other. This category grouping has only a very small portion of data classified as "rare" which will increase model performance and stability by reducing outliers and reducing the possibility of model overfitting from infrequent values in the Marital_Status. After categorizing, Education, and Marital_Status were converted into numerical format using one hot encoded using drop_first=True, which avoids the dummy variables and allows us to make these variables compatible within machine learning models. The transformed values was a much cleaner, useful, and informative dataset for clustering and classification purposes.

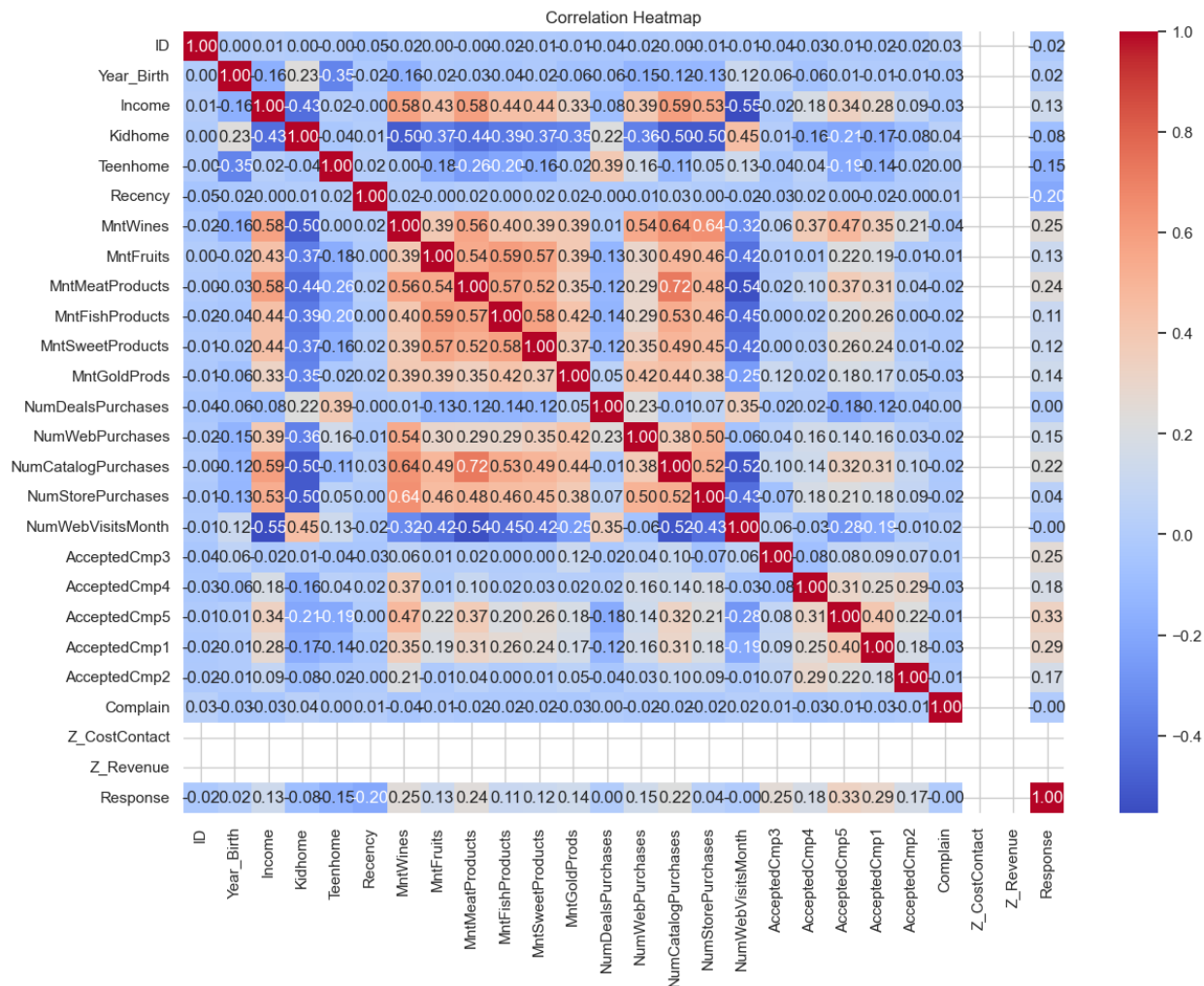
As part of one-hot encoding, Education, and Marital_Status categorical variables were transformed into binary or dummy variables. This allows the categorical variables to be utilized for machine learning algorithms. Using drop_first=True avoids the dummy variable trap related to multicollinearity in regressions. The first category in each variable is dropped and used as the reference. In Education the 2n Cycle level was omitted and is the reference category and the

dummy variables are encoded based on the remaining categories Basic, Graduation, Master, and PhD. Interpretability is maintained while keeping the data numerically stable for modeling.

Correlation Heatmap

Figure 4

Correlation Heatmap



Note: A correlation heatmap represents the relationships between numerical features in the dataset. The heatmap provides a useful in discovering multicollinearity, and to identify redundant variables. Features were dropped after analysis.

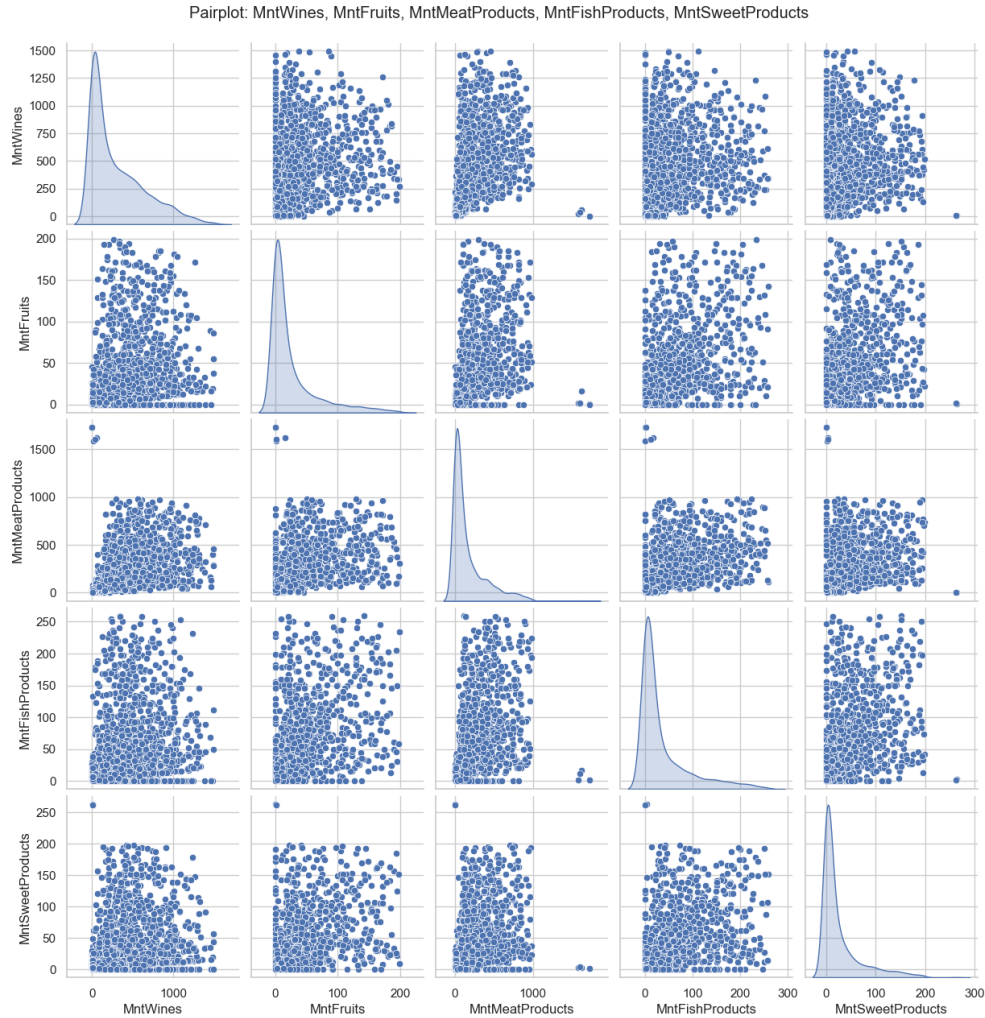
The heatmap shows coefficients across all pairs of numerical features ranging from -1 for strong negative correlation to +1 for strong positive correlation. Strong positive correlations are observed among the product and spending variables MntWines, MntMeatProducts, and MntGoldProds indicating that there is overlap in customer spending behavior among these products. In addition, it can be seen that moderately correlated variables such as NumCatalogPurchases and NumStorePurchases ($r \approx 0.52$), which means that customer engagement tracking is somewhat redundant. Finally, synthetic variables such as Z_CostContact and Z_Revenue showed no meaningful correlation.

Based on the analysis, several features were dropped for model development. Like ID was dropped as it serves as a unique identifier, adds no predictive value. Year_Birth was replaced by Age, which is more interpretable. Kidhome and Teenhome were already captured in new Children variable. Dt_Customer was converted to recency, more relevant for modeling. And Z_CostContact and Z_Revenue are constant or uninformative variables with no significant correlation or variance.

Pairplot of Numerical Distribution

Figure 5

Pairplot of Product Spending Behavior



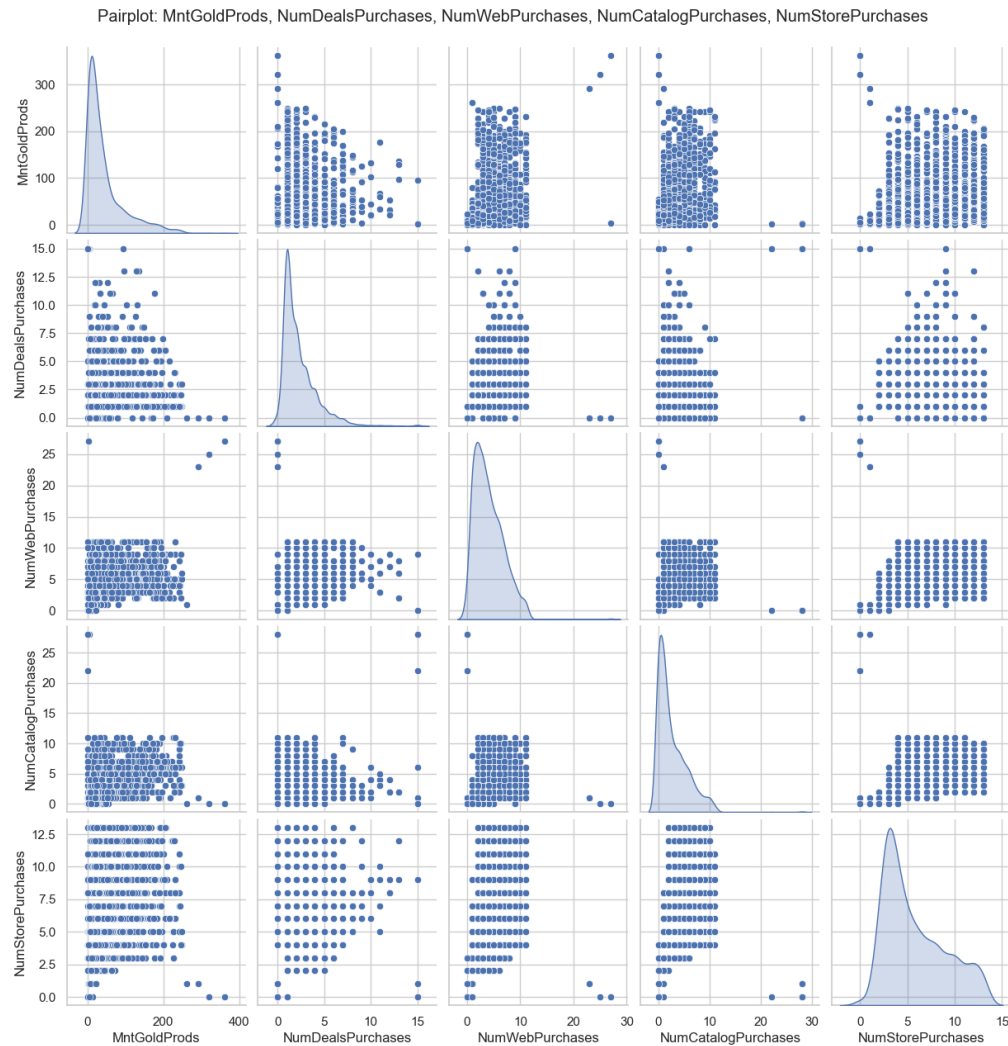
Note: The pairplot explores the associations between the five main product spending variables which are MntWines, MntFruits, MntMeatProducts, MntFishProducts, and MntSweetProducts.

The diagonal plots reveal the five variables are all right-skewed, suggesting simply that most customers spend very little and some few relatively high-spending customers create the long tails. The scatterplots below the diagonal indicate mild to moderate positive relationships between some product categories, especially between MntWines and MntMeatProducts, suggesting similar purchasing behavior. The relationships are not strictly linear though, which also supports the possibility of non-linear models. This visual examination of the data was

helpful for confirming the existence of outliers and clusters of spending which would later be useful for segmentation in later product predictions modeling.

Figure 6

Pairplot of Purchase Behavior Across Channels

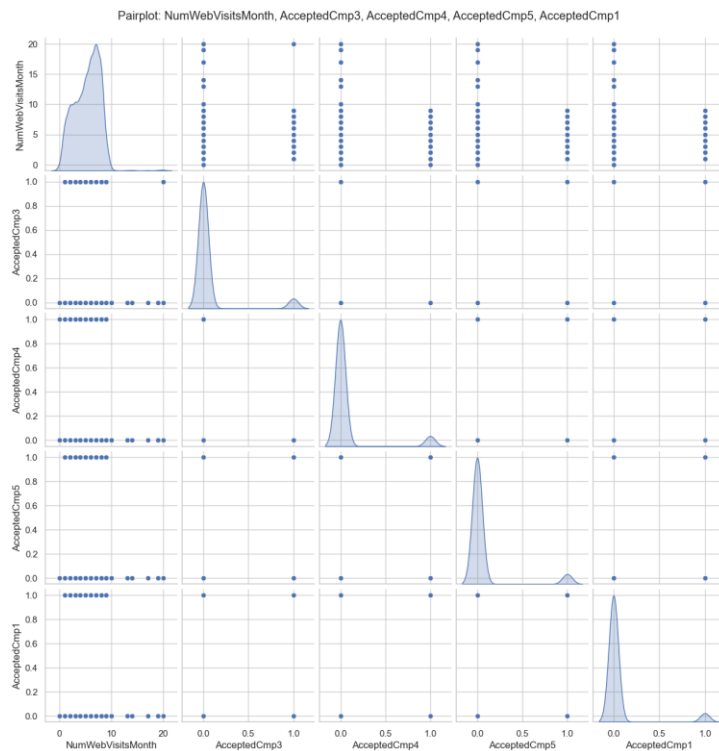


Note: This pairplot shows the relationships MntGoldProds (gold product spending) had with the five purchasing behavior metrics such as NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, and NumStorePurchases.

The distributions on the diagonal suggest that most features are right-skewed, meaning that the customers tend to make less purchases on all channels, and there are just a few highly active shoppers. There is also evidence of fairly moderate clustering behavior around NumCatalogPurchases and NumStorePurchases, indicating some overlap of consumer preferences towards engaging in a physical channel. NumWebPurchases shows very dispersed behavior, with extreme outliers. The scatterplots of the pairplot also support grid-like patterns, which validates the discrete nature of these features counts as an integers. This analysis is valuable in identifying behavioral purchasing segments and supporting classifiers and downstream clustering.

Figure 7

Pairplot of Web Activity and Campaign Responses

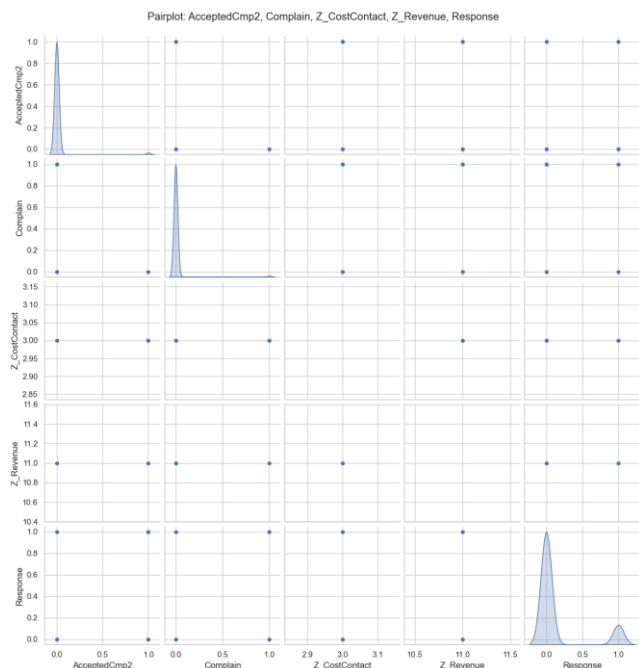


Note: This pairplot examines the relationship between the number of monthly visits to a website NumWebVisitsMonth and the number of marketing campaign responses across four different campaigns AcceptedCmp1 to AcceptedCmp5.

The diagonal sub-plots reveal NumWebVisitsMonth is slightly right skewed, while there is binary imbalance with the campaign acceptance variables, people mostly did not respond (0), and only a few responded positively (1). The scatterplots illustrate little correlation with web visits and campaign responses, which indicates that just because people are active on the site does not mean they will respond to campaigns. Furthermore, not having strong linear relationships between the individual campaign variables suggests customers' response behavior is difference in nature and specific to the campaign. This visual explains that while marketing acceptance is sparse, most people do not accept or responded to campaigns, it is informative enough to use as a signal of customer behavior for segmentation and predictive analytics.

Figure 8

Pairplot of Sparse and Low-Variance Features



Note: This pairplot shows the AcceptedCmp2, Complain, Z_CostContact, Z_Revenue and Response. Looking at the diagonal plots we see that most of these variables are binary and clearly imbalanced, as is the case for AcceptedCmp2 and Complain which have the majority of it being 0's.

Z_CostContact and Z_Revenue appear to have no variance as they appear as a single vertical line, that is, they are constant across all observations and do not add any information. Although Response is slightly imbalanced, it still shows a little distribution and was kept for modeling. Both Z_CostContact and Z_Revenue had no variance and were dropped entirely to limit model processing and noise.

Feature Scaling Using Standardization

After the feature engineering and one-hot encoded was done, our dataset had expanded to include a total of 32 features which at this point included numerical variables such as Income, Recency, and product spending amounts as well as various binary columns for each of the

Education and Marital_Status features after one-hot encoding. Next to standardize the data and feature with the StandardScaler which is found in scikit-learn (sklearn). StandardScaler is used to standardize features so that their distribution has a mean of 0 and a standard deviation of 1. This was done so that no one feature could dominate due to magnitude as many modeling algorithms such as K-Means, logistic regression, neural networks, etc. and expect data to be on similar scales and distributions.

The scaled dataset containing standardized values across all features had no serious outliers anymore. Continuous variables where there was a significant range numerically like. Income, MntWines, Total_Spent were centered around zero, where the one-hot encoded binary variables also scaled to the same level. The scaling helped both improve the interpretability of the model, and the scale helps the algorithm apply equal weighting fairly across unconditioned input dimensions.

Unsupervised Learning: Customer Segmentation

Customer segmentation was done using multiple clustering algorithms to identify different customer segments and to check for underlying behavioral patterns. K-Means was used for partition clustering and combined with Agglomerative Hierarchical Clustering and DBSCAN. PCA was done to improve the results, which preserved 95% of the variance in the data. PCA not only increases function speed, but it also decreases the influence of redundant or noise features.

Kmeans Clustering. The clustering process was started with K-Means, one of the most common clustering algorithms. To decide the optimal number of clusters, for defining (K), Elbow Method was used, where the plot was created of the Within-Cluster Sum of Squares (WCSS) against K values ranging from 2 to 10. WCSS continued to decline as we increased K, however, the elbow point started around K=2, the appropriate number of clusters suggested. To

further confirm our findings, we used different internal clustering metrics, such as Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. These metrics all pointed to $K=2$ as the best cluster between within-cluster cohesion and separation. After fitting the model at $K=2$, we identified two distinct customer segments. The first cluster indicated lower income and product spend, while the second cluster consisted of much higher-income customers, who also engaged significantly more with store, catalog, and online channels. The two groups also varied in their responses to marketing campaigns, as well as how frequently they purchased across the study time period.

Agglomerative Hierarchical Clustering. This model was used to further examine the model. Metric evaluation was done across varying numbers (K). K ranges from $K=2$ to $K=10$, and were evaluated using four metrics, WCSS, Silhouette Score, Calinski-Harabasz index, and Davies-Bouldin Score. Gradual improvement occurred in the clustering quality as K varied, $K=10$ was the measure using which a qualified trade-off occurred across the four evaluation measures. As a result, $K=10$ was selected as the number of clusters. The sets of clusters afford a better resolution of customer behaviours, such that we could identify a number of niche profiles such as, high-value multichannel purchasers, purchasers with responder offers, price purchasers, and low-engagement customers. The resulting segmentation outcome included more detailed resolution than K-Means, and it created contextual data for the subsequent predictive modeling, and strategy development.

DBSCAN. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was used as it conducts clustering based on density with the ability to use noise. It is different from K-Means and Agglomerative Clustering, DBSCAN does not require the number of clusters to be defined before conducting the analysis. Also, it can work with oddly shaped clusters, non-

spherical forms or round objects as well as noise points, which K-Means cannot. Initially, it detected 19 clusters and 1120 noise points with default settings, then targeted the `eps` variable and assessed the `min_samples`. Changing `eps` from 2.0 (19 clusters and 1120 noise points) to 3.0 (11 clusters and 816 noise points) to 4.0 (9 clusters and 404 points) to 5.0 (7 clusters and 169 points) to 6.0 (6 clusters and 44 points). With this entire analysis in mind, `eps = 5.0` was selected since this provided a usable structure of clusters with a manageable amount of noise. Overall, while it highlighted outliers and dense pockets of behavior, there were more challenges with distinct interpretation of clusters in comparison to K-Means or Agglomerative outputs. Therefore, DBSCAN was retained as a secondary clustering method only, and to help settle issues for possible boundary cases and behaviors of customers potentially considered to be within the optimal range in their respective clusters.

This multi-model clustering approach allowed to segment customers at varying levels of granularity and complexity. Cluster labels were added to the main dataset to create additional features in the supervised learning models that followed.

Principal Component Analysis. To improve the clustering performance and reduce computations, Principal Component Analysis (PCA) was performed after developing baseline models to improve the results. PCA generates a lower-dimensional map from the scaled dataset by selecting independent components with the most variance. Thus, PCA transforms the original dimensional feature space into a lower solitary dimensional space where it can be assessed the information of the original data across principal components. The first 10 principal components were assessed by how much variation was captured. The results indicated that the first component alone was able to capture about 22.97% of the variation, with the first 10 components captured a total of 64% of the variation. Thus, a substantial amount of the original information in

the dataset could be maintained using a relatively low principal component number and PCA was a good decision as a dimensionality reduction method.

Following PCA, both K-Means and Agglomerative Clustering were reapplied on the transformed data. K=2 was previously the selected optimal cluster size for K-Means using the Elbow Method, but with PCA applied to it, it founded a new structure to the data that enabled the model to uncover more diverse and meaningful new groupings. In this instance, K-Means created 10 clusters over the PCA-reduced dataset, unveiling richer segmentation. The discovery of cluster profiles revealed a number of behavioral and demographic differences among clients, including income levels, product spend, campaign responsiveness, level of education and marital status. A similar outcome was identified in Agglomerative Clustering, wherein again, metric-based evaluation determined again that we should create 10 clusters. In this context, the new clusters had overlapping customer groups but were uniquely nuanced from the original clusters created without applying PCA, suggesting that dimensionality reduction aided with uncovering some hidden behavioral characteristics from the data.

However, DBSCAN's didn't work out on the PCA transformed dataset. While K-Means and Agglomerative Clustering benefited from linear transformation of features, DBSCAN did not. The initial DBSCAN run resulted in finding only one meaningful cluster that didn't include noise, despite the hypertuning. It was experimented with a wide range of eps values, from 0.3 to 10.0, to check cluster density, but the results consistently failed to produce multiple meaningful groupings. For example, eps=0.3 yielded 12 clusters but a very long list of noise points, eps=0.5 yielded a number of clusters of three. Anything more than eps=0.7 yielded repeatably a number of cluster results of 1 cluster or all was noise - regardless of the eps parameterization value. These results denote that PCA found no dense areas and hence with the nature of this reduced

data there was no non-linear separations to form meaningful clusters. Therefore, in some way, DBSCAN is not suitable to use on the PCA reduced dataset which is why it was dropped from the final segmentation analysis.

It was observed that PCA was very useful in improving segmentation with K-Means and Agglomerative Clustering to uncover deeper insight about the customers. On the other side, it diminished the utility of DBSCAN, which relies on variations in local densities that were probably neutralized from the linear transformation step.

Supervised Learning: Product Purchase Prediction

In the study's supervised learning stage, to predict the next product a customer would purchase. To accomplish this, five different classification models such as Logistic Regression, k-Nearest Neighbors (k-NN), Decision Tree, Random Forest, and XGBoost were created. These models were fitted to pre-processed features and evaluated using five different metrics which are Accuracy, Precision, Recall, F1-Score. These metrics offered a broader picture of each model's performance in the context of an imbalanced multiclass data problem. The Random Forest and XGBoost baseline models gave the best results overall, meaning the models have the capacity to discover complex decision boundaries in customer behavior.

HyperParameter Tuning. To further enhance performance, hyperparameter tuning was implemented with GridSearchCV, including 3-fold cross-validation, to guarantee that all product classes were optimized to some degree. For each model, it was established a parameter grid and supplied it to the grid search loop that saved the best estimator upon completion. XGBoost required some special handling due to needing the target labels to be encoded.

Initially, the model's performance did not change much after applying hyperparameter tuning, which means that it was not about the model complexity. After further investigation, trying different things, it was discovered that the model performance was so low because there was a significant class imbalance with the target variable, it showed TargetProduct such as wine (0) - 0.687054, meat (2) - 0.197768, gold (5) - 0.075000, fish (3) - 0.026339, sweets (4) - 0.007143, fruits (1) - 0.006696. On the basis of this SMOTE technique was used to rectify this imbalance when training the model.

Synthetic Minority Over-sampling Technique (SMOTE). In order to address the imbalance in the target variable the Synthetic Minority Over-sampling Technique (SMOTE) was applied during the training process of the models. The target variable represents the product category a customer is most likely to purchase and showed considerable imbalance which means that the majority of customers were concentrated in one or two classes and several of the classes had far too few instances. In these types of situations, the model can be biased to predict the dominant classes and ignore the minority classes. SMOTE addresses class imbalance by synthetically creating new samples for the minority classes, rather than duplicating existing instances, SMOTE synthesizes new instances for the minority classes through interpolation between instances. This resulted in a training set data which is less biased towards a particular product class, which allowed model to learn patterns in the classes more effectively irrespective of their imbalance and reduce the risk of making biased predictions. Importantly, SMOTE was only applied to the training data to avoid information leakage once the train-test split occurred and protecting the integrity of model evaluation, so that the model is not biased and can be used in real world scenarios. Class distribution after SMOTE looked like this TargetProduct, 0 - 1539, 2 - 1539, 1 - 1539, 3 - 1539, 5 - 1539, 4 - 1539.

This supervised modeling, integrated with the unsupervised segmentation, enabled the system to be more relevant. Predictions were personalized for each customer and informed specifically by the behavioral clusters they belong to. The overlap between customer identity segment and predicted intent product established an end-to-end pipeline to target strategic by retail and marketing.

Results

Unsupervised Learning: Customer Segmentation

Baseline Model Results.

Table 3

Baseline Model Results

Model	Silhouette	Calinski-Harabasz	Davies-Bouldin
Kmeans (K=2)	0.214755	489.949192	1.994479
Agglomerative (K=10)	0.148104	165.233367	1.746453
DBSCAN	0.212461	56.786617	1.112297

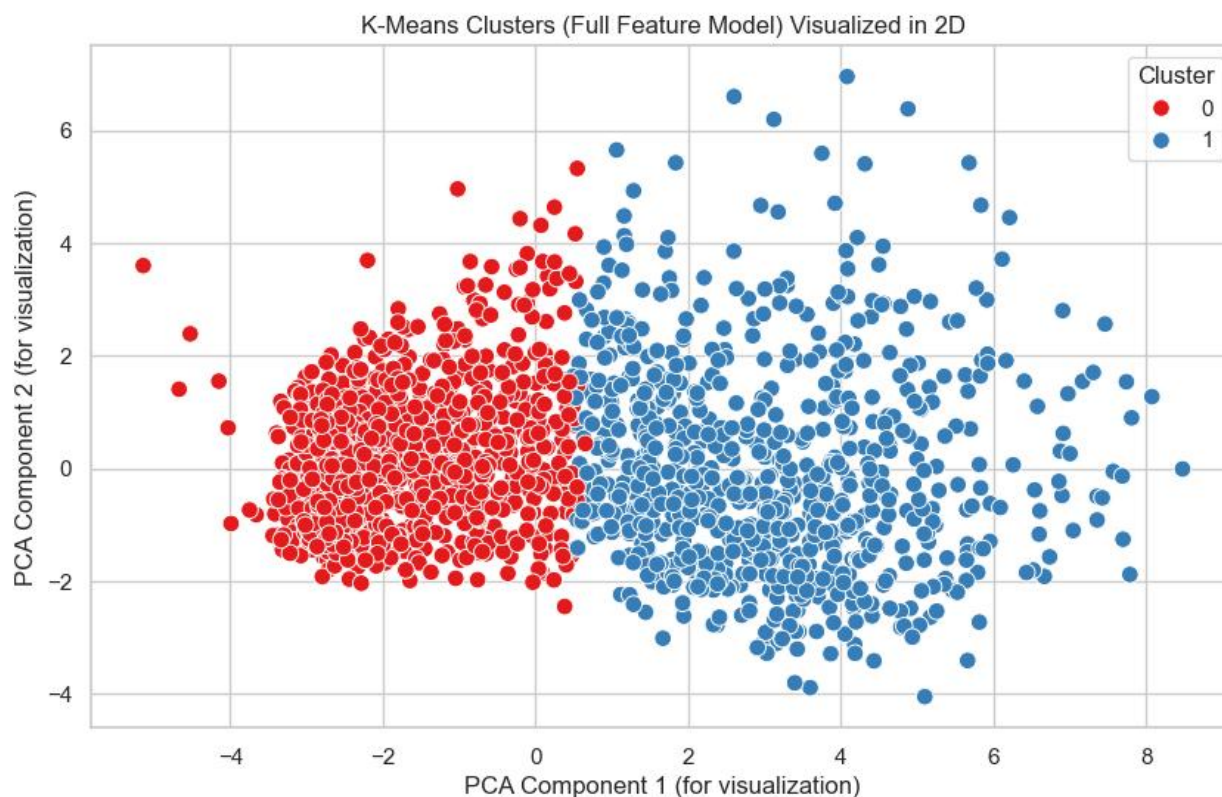
Note: Table shows how the clustering models performed using different evaluation metrics. K-Means (K=2) had the highest Calinski-Harabasz score and lowest Davies-Bouldin index of all mentioned algorithms, which indicates compact and distinctly-separated clusters.

K-Means also had the highest silhouette score. Agglomerative clustering (K=10) did yield more detailed groupings and even with a more granular level of clustering, it didn't produce the highest silhouette score nor a higher Calinski-Harabasz score, so the resulting clusters were slightly less well defined but still produced distinct clustering. While DBSCAN produced a silhouette score similar to K-Means' and had a low Davies-Bouldin index suggesting it had good intra-cluster similarity, its overall clustering performance was quite limited by the fact that it had

poor separation between clusters. K-Means is the best-performing model overall based on the combination of clustering evaluation metrics and practical interpretability.

Figure 9

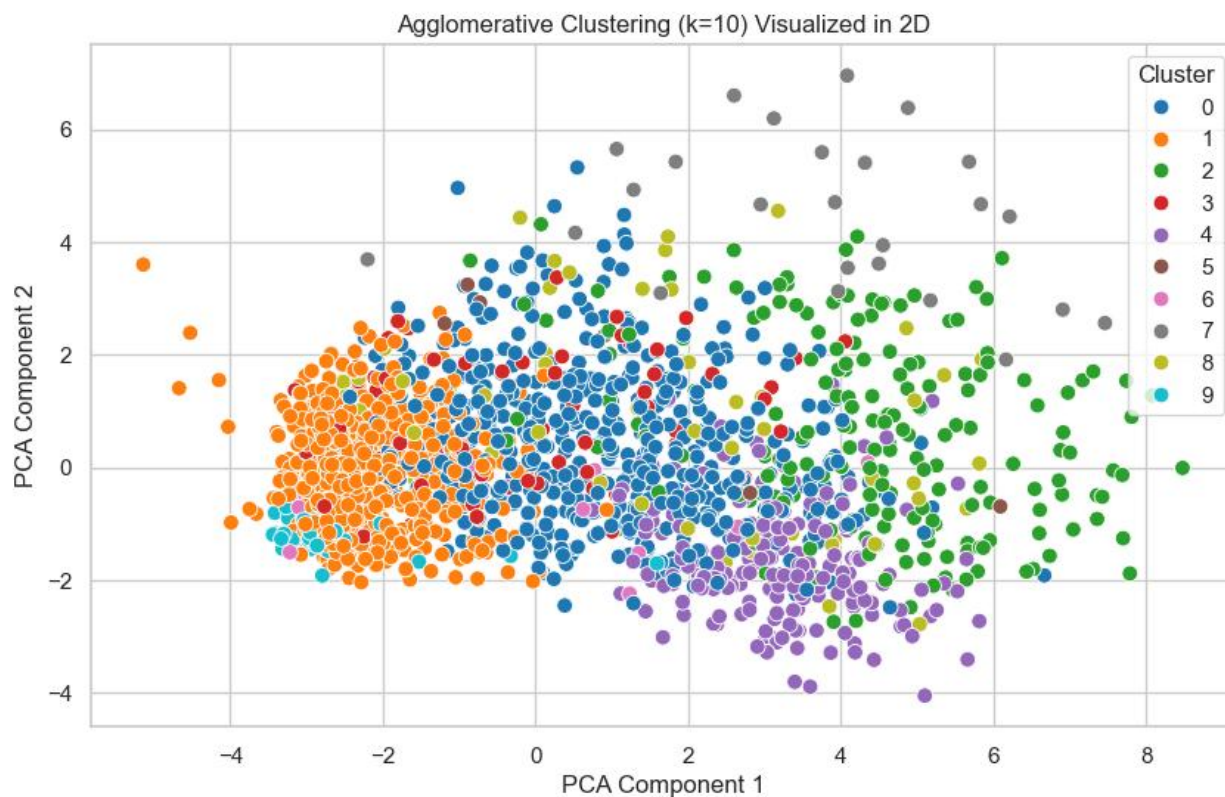
Kmeans (K=2) baseline model



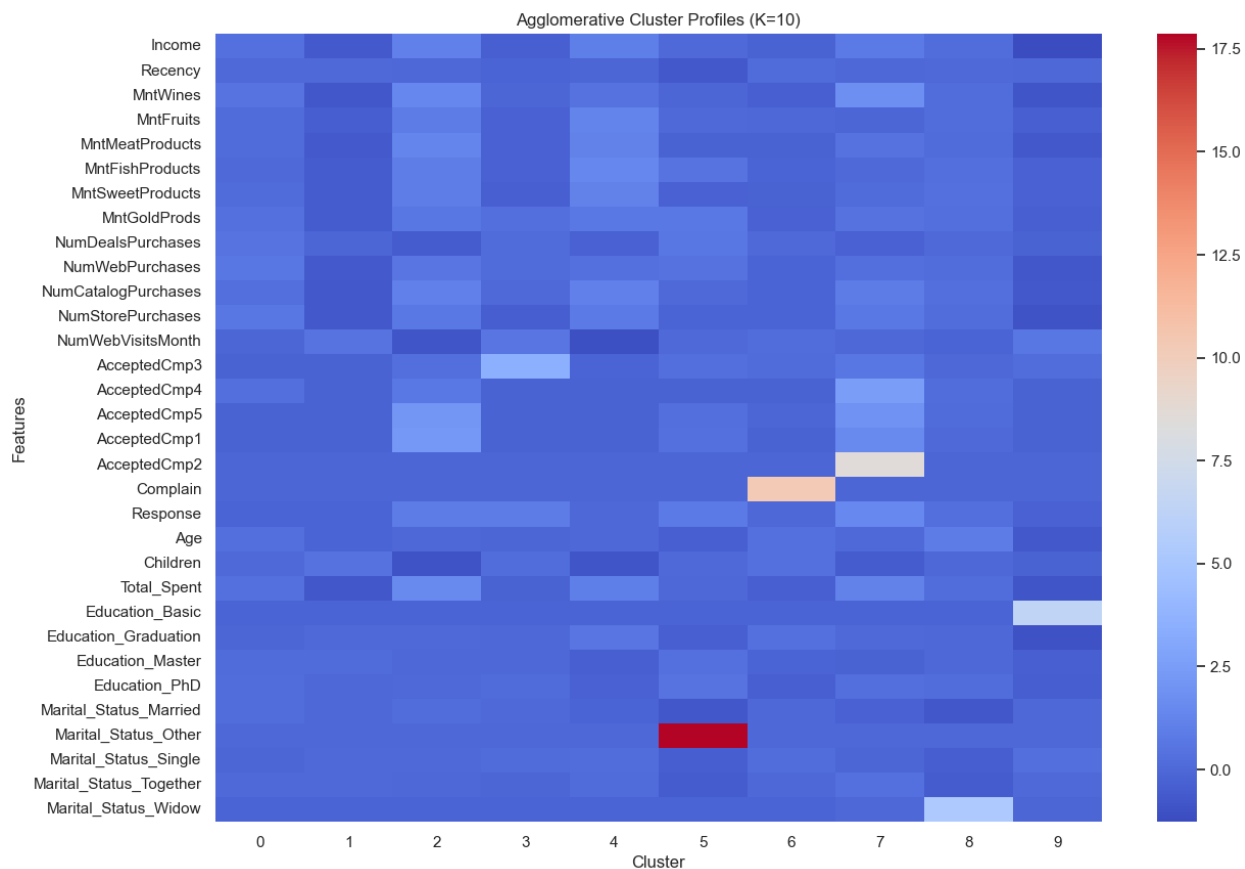
Note: It is important to note that the visualization of the K-Means clusters presented above is in 2D using PCA components for distance only, and the 2 clusters are well-separated based on where clusters are plotted with minimal overlap between the red and blue clusters. This clustering structure demonstrates distinction across the full set of features, and thus it was quite clear there was meaningful segmentation before any dimensionality reduction. Clusters were Lower-income, low-engagement buyers and Higher-income, highly engaged premium customers.

Figure 10

Agglomerative (K=10) baseline model

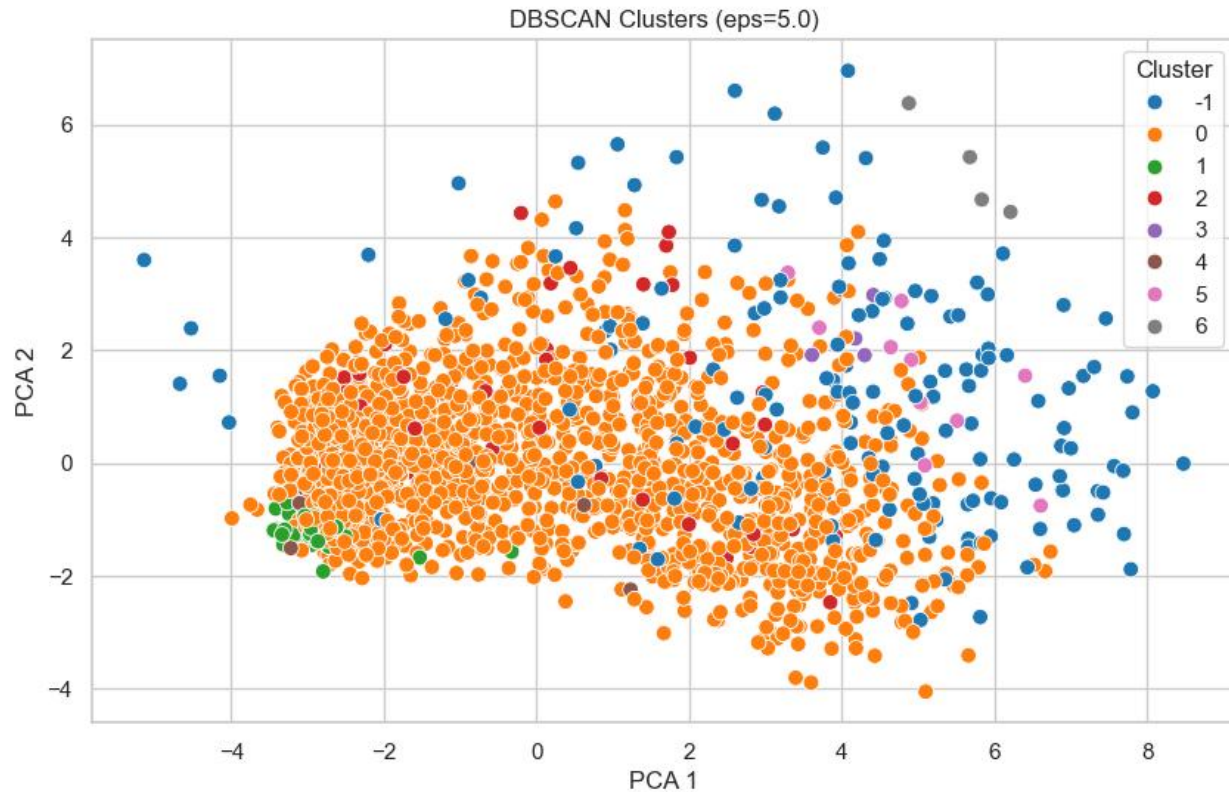


Note: This graphic shows the Agglomerative Clustering output (K=10) in two dimensions (not preserved distance; using normalize PCA) with overlap as some clusters overlap. The data is divided into ten different clusters which show a slightly more complex separation than standard K-Means, although a few clusters are very mixed and densely packed in areas as well which is reflective of the hierarchical structure in the model. Clusters type were Budget-conscious minimal spenders, High-income bulk buyers, Selective campaign responders, Digitally active moderate spenders, Loyal offline buyers.

Figure 11*Agglomerative Clustering profiles Heatmap*

Note: This heatmap displays the Agglomerative Clustering profiles (K=10) for the hierarchy before PCA. Each column represents one of the clusters, while the rows correspond to different feature intensities: income, spending on products, number of responses to campaign, demographics, etc. The color spectrum is a visualization of the variation of a cluster in these characteristics - for example Cluster 5 has high "Education_PhD" and Cluster 6 has a strong response to campaigns. This visualisation will be useful in identifying unique groups of customers based on their behaviours and characteristics.

Figure 12*DBSCAN baseline model*



Note: The DBSCAN clustering visualization shown above indicates the outcome of density-based clustering with epsilon (eps) = 5.0. Most points belong to Cluster 0 (orange) indicating a large core cluster. DBSCAN has also identified six smaller clusters and a set of noise points (Cluster -1 in blue), which are scattered due to low density and therefore, do not belong to any cluster. This clustering has shown that DBSCAN managed to find some local variations, but still struggled outlining clear cluster boundaries as a result of the data being so overlapped and with density variations, even after a PCA transformation.

Model Results after PCA.

Table 4

Model Results after PCA

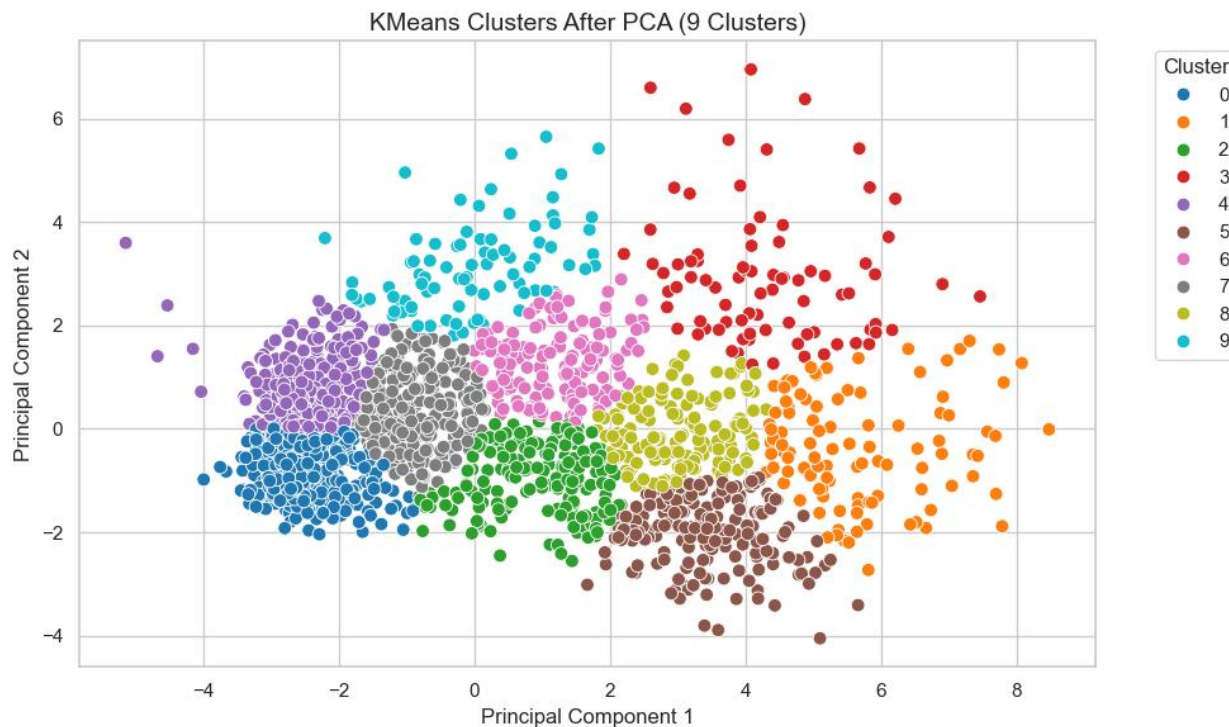
Model	Silhouette	Calinski-Harabasz	Davies-Bouldin
KMeans (PCA)	0.376622	2420.43628	0.844765
Agglomerative (PCA)	0.306698	2046.964118	0.910211

DBSCAN (PCA)	NaN	NaN	NaN
--------------	-----	-----	-----

Note: After PCA had been applied, the clustering performance improved across both K-Means and Agglomerative clustering models. K-Means achieved the highest silhouette score (0.3766) as well as the highest Calinski-Harabasz index (2420.43) with the lowest Davies-Bouldin index (0.8448) which indicates maximum possible intra-cluster compactness and minimum inter-cluster separation. Agglomerative clustering was again similar to K-Means but slightly lost out on all evaluation metrics when compared to K-Means results. DBSCAN was unable to provide valid clusters post PCA resulting in no evaluation metrics to be provided. The above results provide confidence that when combining K-Means and PCA as well the K-Means clustering with PCA provides the most compact and well-separated clusters and was the best model for customer segmentations for this problem context.

Figure 13

Kmeans (K=10) after PCA



Note: The visualization shows the K-Means clustering after the PCA transformation. The model was configured to find 10 clusters, although the title represents 9. The PCA reduced the dataset into 2 principal components to improve visualization. The clusters look to be relatively separated and have clear definable boundaries and the groups appear to be more compact and spherical than the pre-PCA version. This suggests the PCA transformation has removed some noise and redundant features that improved the overall clustering structure.

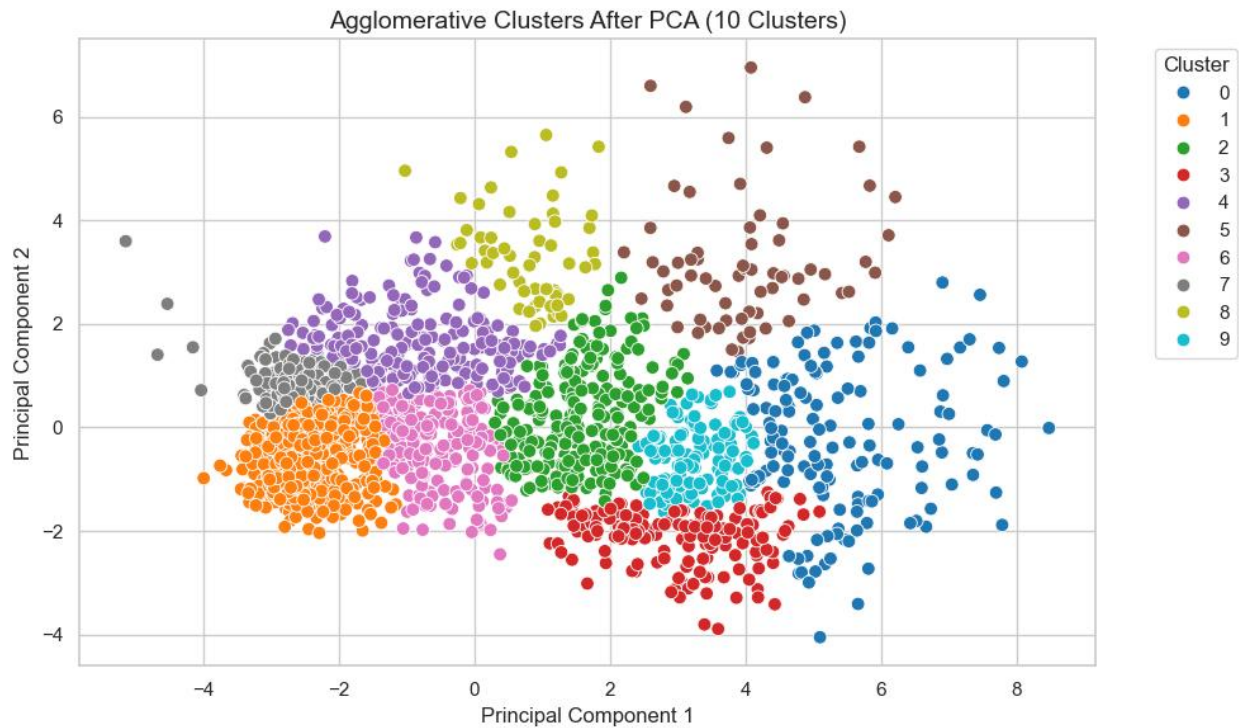
The customer segments generated from the KMeans clustering analysis led to 10 unique clusters with interesting demographic and behavioral characteristics. Cluster 1 is a group of high income high-spending customers, as identified by highest normalized values for income, and a higher normalized spending value across product categories as diverse as wines, fruits, meats, fish, sweets and gold products. This group is highly responsive to the online world, they often buy online and are willing to engage with deals, thus indicating both a more highly developed digital presence and brand sensitivity. Next is Cluster 0, a low-income, low-spending group that

favors conservative shopping, with negative scores on all the spending variable, and very limited interaction with promotional deals and online purchasing. Similar to Cluster 1, Cluster 3 is characterized by high spending on wines driven by a group of affluent wine appreciators, however across all other product types they only have moderate or low engagement as a consumer group. While they may fall into the affluent cluster they may have a more niche lifestyle segment criteria the characters their shopping activity. Cluster 5 could be identified as balanced family spenders, who active engagement is suggested by strong purchasing behavior in food categories of fruits and meats, along with a modest but reasonable income level. Finally, Cluster 9 is unique due to having extremely high online purchasing behavior and high acceptance of deals, but modest income and spending values, depending on the product type, and are likely a segment driven largely by typical promotional activity and e-commerce.

Clusters 6 and 8 also represent above-average earners with moderate to high product interest but varied patterns, Cluster 8 is more aligned with meat and fish purchases while Cluster 6 is more in wines and web engagement. Cluster 2 includes the middle-class cautious shoppers, with average above-median income and some engagement across products and purchase channels. Cluster 4 is a very disengaged consumer with low scores on expenditure spending, income, and online activity, likely the older consumer, or price-sensitive consumer. Finally, Cluster 7 is a neutral group with average values across majority of metrics, not especially inclined toward any product or platform. These profiles may help in product directing or targeting, promotional strategies, and campaign personalization to align promotional messaging with the needs and preferences of each cluster.

Figure 14

Agglomerative (K=10) after PCA



Note: The figure shows Agglomerative Clustering post PCA which identified 10 clusters in reduced 2D space. The clusters are visually more distinct and compact than in the full feature space, confirming that PCA improved the clarity of group boundaries. Each cluster has a characteristic shape and density, allowing for better interpretation of customer groups. The improvements are more visual than based on metrics, stressing PCA's ability to assist in accounting for cluster differences.

PCA provides moderate improvement in cluster performance like more separation and compact clusters. For example, K-Means registered an increase in silhouette score, and the Calinski-Harabasz index was significantly higher, which indicates that clusters were defined more clearly. While agglomerative Clustering benefited from PCA, the improvements were not as substantial. PCA reduced noise and computational resources, but the original cluster structure was relatively consistent. While the value of improvement provided by PCA was useful, it is not

revolutionary. PCA suggests that useful patterns already exist in the original feature space and K-means and Agglomerative Clustering just refined those patterns versus changing the segmentation result.

Supervised Learning: Product Purchase Prediction

Baseline Model Results.

Table 5

Baseline Model Performance Comparison

Model Name	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score
Logistic Regression	0.72	0.63	0.72	0.64
k-NN	0.72	0.68	0.72	0.69
Decision Tree	0.70	0.69	0.70	0.69
Random Forest	0.73	0.72	0.73	0.72
XGBoost	0.74	0.72	0.74	0.72

Note: This table provides a summary of the baseline performance of the five models using accuracy, weighted precision, weighted recall, and weighted F1-score. Weighted metrics was used to address class imbalance and to provide an overall view of the model performance across product categories. XGBoost is the best model.

Amongst the five baseline models, XGBoost achieved the highest accuracy of 0.74, and also almost in line with Random Forest's weighted F1-score of 0.72, suggesting it showed strong performance overall, with consistency across classes, despite the imbalanced dataset. Random Forest, because it is an ensemble method, also handled class imbalance well, and reduced the possibility of overfitting by capturing interactions and non-linear relationships between features. Decision Tree accuracy had an accuracy of 0.70 which was lower comparatively, but had balanced precision and recall. Although k-NN also produced decent results and showed accuracy

of 0.72, it was more likely that k-NN accuracy suffered from the added value and complexity to the multiclass problem. Although both k-NN and Logistic Regression achieved a 0.72 accuracy, LR had the lowest precision, recall and F1-score, indicating that it likely had limitations modelling underlying complex, and or non-linear, interactions in the context of multiclass. Overall, it can be said that ensemble methods were consistently more effective than any of the simple models, and these were rated as the most promising models to optimize.

Model Results after SMOTE

Table 6

Model Performance Metrics After Applying SMOTE

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score
Logistic Regression	0.73	0.67	0.73	0.67
k-NN	0.71	0.67	0.71	0.68
Decision Tree	0.70	0.70	0.70	0.70
Random Forest	0.75	0.72	0.75	0.73
XGBoost	0.77	0.75	0.77	0.75

Note: The table shows the performance measures of five classification models after using SMOTE to address class imbalance in the target variable. Slight improvements in accuracy and weighted F1-score, but the results show that SMOTE did not greatly improve the models' predictive capability for each of the classes.

All five models showed improvement in overall performance after SMOTE was applied to the original imbalanced target classes. However, XGBoost was the top performing model, with an overall accuracy of 0.77 and weighted F1-score of 0.75. SMOTE improved the model's ability to uncover more complex patterns, enabling it to learn better from the balanced training data. Similarly, Random Forest was also able to generate strong results with an overall accuracy of

0.75 and weighted F1-score of 0.73. This model uses an ensemble structure often capturing more unique and relevant data and was sensitive to noisy data while trying to identify distinct unseen structures. Decision Tree, Logistic Regression, and k-NN showed modest, but consistent increases in overall performance from the rest of the models, where Logistic Regression achieved an accuracy of 0.73 and Decision Tree maintained tried to maintain stable precision, recall, and F1-scores of 0.70. The overall results suggest SMOTE has allowed the models to generalize better across all classes, with the most improvement with the two minor classes.

Comparing with the baseline models, SMOTE influence is clearly visible. XGBoost's accuracy improved from 0.74 to 0.77 and increased its F1-score from 0.72 to 0.75, meaning the model was better positioned to learn minority class patterns after balancing. Random Forest saw an accuracy increase from 0.73 to 0.75 and F1 score increase from 0.72 to 0.73. Logistic Regression, which lagged initially for precision, showed an increase from 0.63 to 0.67, which suggests some improvement in being able to handle class distributions. The improvements seen across models show that using SMOTE to counteract the class imbalance provides fairer, more reliable predictions.

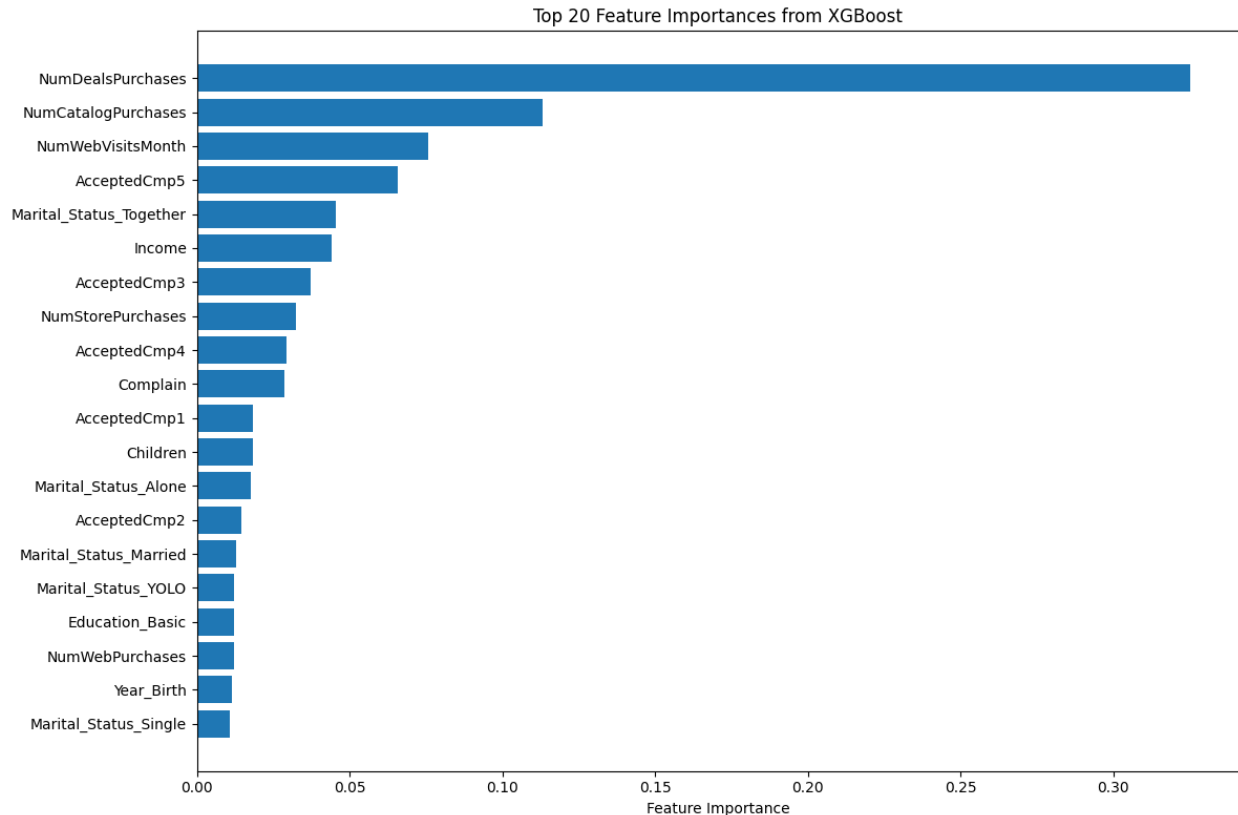
Therefore, even with a balanced training set, overall performance did not improve. The models continued to experience difficulty generalizing across all classes, which suggests that class balance alone was not enough to sufficiently improve model learning or predictive performance in this case. These results suggest that while SMOTE addressed the imbalance problem, it did not dramatically change the model predictions. One possible reason for this could have been that the features themselves, even after SMOTE, features were distinctive enough to allow models to confidently predict all classes. Therefore, while SMOTE improved fairness in

learning across all classes, it did not provide a dramatic improvement in prediction quality with further feature engineering or modeling options needed.

Feature Importance

Figure 15

Feature Importance from XGBoost



Note: This bar chart presents the top 20 features ranked by their importance scores from the XGBoost model. The NumDealsPurchases and NumCatalogPurchases features turned out to be the most significant factors in predicting the target product which indicates the importance of customer's behavior in regards to purchasing decisions regarding products. The other significant features include online engagement NumWebVisitsMonth and some responses to marketing campaigns which show the importance of behavioral factors and promotional factors in the classification task.

Dashboard of Results Predicted

Table 7

Dashboard representing product predicted customer segment wise

Dt_customer	Predicted Product	Customer Segment
961	MntWines	2
229	MntMeatProducts	2
1093	MntWines	0
427	MntWines	3
1650	MntFishProducts	2
543	MntWines	3
1801	MntWines	1
1708	MntWines	2
994	MntWines	2
976	MntWines	1

Note: The above table shows the output of the dashboard reflects the predictions for individual customers around product purchases in a collated view with their unique ID (Dt_customer), sustainable product they would be more likely to purchase (MntWines, MntMeatProducts, MntFishProducts), and the respective customer segment they are identified as (from clustering in unsupervised learning).

This hybrid approach of clustering through unsupervised learning with the prediction of products through supervised learning provides a robust, data-driven personalization approach. First, this model identifies segments of consumers, then predictions through those segments more likely, with better accuracy. For example, Segment 2 appears very regularly in this dashboard and is particularly strong towards wine and fish in product affinity, which would have gone unnoticed without clustering. This prescriptive information provides the connection beyond just what a consumer is likely to purchase next, as it also provides the correlations between the underlying

customer type, this allows businesses to improve marketing, cross-selling, and inventory planning. By hybridizing behavioral segmentation with predictive modeling as illustrated in this report, decision-making is improved to enable more precise and actionable recommendations.

Discussion

The unsupervised learning phase of the project aimed to identify latent customer segments, based on demographic and temporal characteristics, using clustering algorithms. K-means and agglomerative clustering methods both generated 10 customer clusters after PCA transformation completed, ultimately segmented the customer base into clustering profiles identifying low income, low spend customers and high-income loyal buyers, responsiveness to campaigns, and digitally engaged customers. Clustering profile interpretations were confirmed using heatmaps and associated cluster feature analysis. PCA provided modest, yet definitive, contributions to the improvement of interpretability and slight tightening of clusters due in part to the reduction of redundancy across the dataset. Interestingly, DBSCAN ultimately provided an alternative interpretation with regard to which customers were core points and core noise with few niche clusters made up of tightly clustered, highly engaged customers and identified outliers displaying erratic behaviors. That said, it is worth noting that DBSCAN is very sensitive to parameter specification making it unstable depending on dimensions. Overall, clustering allows the construction of a behavioural map of the consumer base that shapes the subsequent approaches to personalization.

In the supervised learning phase, it was a multiclass classification problem, while models were trying to predict the next product a customer would purchase. Five baseline models were created such as logistic regression, k-NN, decision tree, random forest, and XGBoost, after preprocessing the data and imputed used all of the features. Random forest and XGBoost model

consistently achieved better performance than the simpler models. For XGBoost results, it was worth noting that we achieved the highest overall accuracy and F1 when applied SMOTE to handle the class imbalance. However, even after performing hyperparameter optimization through GridSearchCV, the performance of the models did not change, there was just the slightest change. None of the models performed as per the real world expectations, even with SMOTE, there was an improvement but the overall performance was not that great. It was observed that the models were still not picking up on the small differences between class types, not able to distinguish due to imbalance in target product.

One of the most surprising observations of this study was that there almost no improvement in the model performance even after hyperparameter tuning or SMOTE. Under most scenarios, tuning increases the generalization of a model and reduces overfitting, but in this instance, the final metrics remained same as the baseline model. This may shows that the dataset lacked sufficient discriminative power or depth to its features, many consumer choices may rest on factors associated with seasonal demand, marketing offers, or emotional influences not included in the training and testing dataset. Even well-performing models like XGBoost and Random Forest may not have benefited from performance tuning, possibly due to redundant features or an over-representation of a dominant product class like wines. In large part these surprises emphasize the complexity of consumer behavior and the challenges that come with deploying productive models consisting of minimal behavioral variables and demographic variable.

Despite this, the value of having a clear data visualization that could tie in the unsupervised clustering results and the outputs from the supervised predictions really added an interpretive layer to the final deliverable as compared to the earlier analysis. The final dashboard

that visualized the predicted product category preferences by customer segments such as Cluster 2 was low-income, wine-preferring buyers provided valuable insights to the marketing and sales teams. That connection allows for prospective operational downstream activities like suggesting product bundles, time optimizations, or loyalty programs targeted to individual clusters. A cluster group that is highly active both on the web and with catalog purchases may respond less favorably to remarketing and should have online promotions instead, while the cluster group that is most responsive to a targeted campaign through a catalog may require more aggressive efforts for remarketing. It is value from the unsupervised learning outcome as the customer segments were revealed according to nature and the supervised learning predicted each customer segment's next action, a client journey map that is both descriptive and predictive.

Limitations

One of the major limitations of this study is its static dataset. The absence of time-related variables like time since last purchase or buying history did not allow the model to change over time in customer behavior. As a result, it was difficult to get a complete understanding of purchase patterns, trends, or recency effects which are important to understanding customer behavior in real-world retail. The absence of any psychographic variables such as brand preferences, lifestyle choices, or price sensitivity, which are often significant dimensions for product deliberation, also limits the prediction capability, even though demographic and behavioral variables were included. Even after employing SMOTE to address issues of class imbalance, because there were still some challenges in predicting some classes. Also, inconsistent results for clustering methods like DBSCAN and flat levels of model responsiveness to hyperparameter tuning suggest that the dataset and features may not be sufficient enough in scope to support production-grade predictive models.

Recommendations for Retail Companies

Retail businesses can gain new insights from this analysis. Retail companies can segment consumers by clustering to construct persona-based marketing plans that closely match "real" consumer behavior. For example, the segments that received high online engagement and response from the campaign could be targeted to an email campaign. Other segments that preferred to purchase in-store were more likely to respond to in-store discounts or loyalty program cards. The clusters are actionable. As an example, customers in cluster 2 were labeled budget-sensitive consumers who bought affordable products such as wine or meat. There is an opportunity to upsell these consumers, especially through bundle deals. There were also consumer segments that had high-income and low-engagement, these types of segments could be cultivated through premium services, targeting exclusivity campaigns. With each of the clusters combined with the supervised product prediction model, there is the opportunity to be proactive and anticipate consumer needs so the retailer can nurture the customer and identify them as engaged before they slipped into not buying mode. The outcome of the state of incomplete systems store engagement model as discussed in previous sections in a real-time environment would require a significant amount of work for development, however, the analysis presented in this is a great starting point in implementing data-driven solutions to think about the new customer journey planning in modern retail.

Conclusion

This research used a hybrid machine learning process to map out the customer journey in the retail domain by segmenting the customers in an unsupervised manner and subsequently predicting product purchases in a supervised manner. Using clustering techniques to identify

customer segments enabled us to understand the behaviour of customers using profiles based on distinct customer segments such as low-income wine buyers, campaign-responsive digital users, consistently high-spending loyal customers. The feature reduction process using PCA helped to provide more interpretability and reduce dimensional noise, while DBSCAN did not provide additional value and led to limited segmentation because of the sensitivity of the parameter tuning and not having clear clusters after transforming PCA.

In the supervised learning part, five classification models were built to predict what product a customer would most likely buy next. XGBoost was the best model with 0.77 accuracy followed by Random Forest. However, the performance improvement was very slight after hyperparameter tuning. The performance of the models does not seem to reflect deeper nuances of product preference, given that it was unable to adequately capture all information from the features we included. Even after applying SMOTE to counter class imbalance and strengthen generalizability across classes, the models performed slight better but overall it was not that great. However, it can be stated that the behavioral and demographic features available were limited.

Although the model result for supervised learning was not something we can use in real world but still, the combination of clustering along with classification to identify meaningful groups and then predict what customer is going to buy next. This combined approach not only improved business value and confidence of the predictions but also served as a useful reminder of how the understanding of a customer's identity is important for creating an appropriate product recommendation.

While the findings present considerable possibilities for targeted marketing and personalization, given their current non-real-time state, the models are not ready for

operationalization. Future work may include time-series information such as purchase frequency, recency, consumer feedback information, and or contextual information such as promotions and seasonality trends. There is considerable value in considering either neural network-based models, or time-series aware models or DL that may uncover rich predictive patterns. However, this research demonstrates the beginnings of a foundation for developing a machine learning pipeline to understand and predict retail customer behavior.

References

- Bhatia, K., & Gupta, B. P. (2025). Predictive analytics for customer behavior and sales forecasting in retail. *International Journal of World of Sciences*.
<https://www.ijwos.com/index.php/home/article/view/10>
- Deldadehasl, M., Karahroodi, H. H., & Haddadian Nekah, P. (2025). Customer clustering and marketing optimization in hospitality: A hybrid data mining and decision-making approach from an emerging economy. *Tourism and Hospitality*, 6(2), 80.
<https://doi.org/10.3390/tourhosp6020080>
- Kabir, M. F., Chowdhury, M. D., Shahiduzzaman, M., & Rana, M. I. C. (2025). Comprehensive customer segmentation and behavior prediction using advanced machine learning and neural network models. *International Journal of Creative Research Thoughts*, 13(1), 478–490.
- Lathifah, S. N., & Azzahra, Z. F. (2025). AI-driven customers segmentation using K-Means clustering. *G-Tech: Jurnal Teknologi Terapan*, 6(1).
<https://ejournal.uniramalang.ac.id/index.php/g-tech/article/view/6202>
- Nasirzonouzi, A. (2025). Leveraging network science for customer segmentation and product recommendation. *Northeast Journal of Complex Systems*, 7(1), Article 3.
<https://orb.binghamton.edu/nejcs/vol7/iss1/3/>
- Patel, R., & Mehta, A. (2024). Hybrid CNN-ANN models for E-commerce analytics. *Deep Learning Applications in Business*, 15(2), 65–80.

Ramkumar, G., Bhuvaneswari, J., Venugopal, S., Kumar, S., Ramasamy, C. K., & Karthick, R.

(2025). Enhancing customer segmentation: RFM analysis and K-Means clustering

implementation. In *Innovations in Management and Technology* (pp. 109–122).

<https://www.taylorfrancis.com/chapters/oa-edit/10.1201/9781003559139-9>

Suh, Y. (2025). Discovering customer segments through interaction behaviors for home

appliance business. *Journal of Big Data*, 12(1), 54. [https://doi.org/10.1186/s40537-025-](https://doi.org/10.1186/s40537-025-01111-y)

01111-y

Awaliyah, D. A., Prasetyo, B., Muzayanah, R., & Lestari, A. D. (2024). Optimizing customer

segmentation in online retail transactions through the implementation of the K-Means

clustering algorithm. *Social Journal of Informatics*, 5(2).

<https://journal.unnes.ac.id/journals/sji/article/view/6137>

Dhanushkodi, K., Bala, A., Kodipyaka, N., & Shreyas, V. (2024). Customer behavior analysis

and predictive modeling in supermarket retail: A comprehensive data mining approach.

IEEE Access. <https://doi.org/10.1109/ACCESS.2024.3407151>

Kaggle Dataset: [https://www.kaggle.com/datasets/vishakhdapat/customer-segmentation-](https://www.kaggle.com/datasets/vishakhdapat/customer-segmentation-clustering)

clustering